MPG No. 13.5.130 $4'm'm$ (4'mm' setting) [Type III, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{xy} = \mathbb{M}^{(1)}_{xy}$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$M_{yzx} = \mathbb{M}^{(1)}_{xyz}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$M_{yzx} = \mathbb{T}^{(1)}_{v}$
$2M_{yzx} = 2\mathbb{T}^{(1)}_{v}$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{yzx} = \mathbb{T}^{(2)}_{v}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$M_{xxxy} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy}$
$M_{zzxy} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy}$
$M_{yzzx} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}^{(3)}_{xy} + 2\mathbb{T}^{\beta(1)}_{z}$$
$$M_{zzxy} = \mathbb{M}^{(3)}_{xy} - 4\mathbb{T}^{\beta(1)}_{z}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}^{(4)}_{xy}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{yzzx} = -\mathbb{M}^{(5)}_{xy} - \mathbb{T}^{\beta(2)}_{z}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyxx} = -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} - 2\mathbb{M}^{(5)}_{xy} + 2\mathbb{T}^{\beta(1)}_{z} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{xxyx} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} + 2\mathbb{M}^{(5)}_{xy} + 2\mathbb{T}^{\beta(1)}_{z} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{zzxy} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + \mathbb{M}^{(3)}_{xy} - 4\mathbb{T}^{\beta(1)}_{z}$$
$$M_{yzzx} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(5)}_{xy} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{yzxz} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(5)}_{xy} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{xyxx} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(3)}_{xy} - 2\mathbb{T}^{\beta(1)}_{z}$$
$$M_{xyzz} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} - \mathbb{M}^{(3)}_{xy} + 4\mathbb{T}^{\beta(1)}_{z}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}^{(4)}_{xy} + \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{yzxz} = -2\mathbb{M}^{(4)}_{xy} + \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyxx} = -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} - 2\mathbb{M}^{(5)}_{xy}$$
$$M_{zzxy} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + \mathbb{M}^{(3)}_{xy}$$
$$M_{xxyx} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} + 2\mathbb{M}^{(5)}_{xy}$$
$$M_{yzzx} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(5)}_{xy}$$
$$M_{xyyy} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(3)}_{xy}$$